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Speech-to-Speech Conversion:
An Approach to Enhance the Speech Intelligibility of Dysarthric Speaker

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ABSTRACT

A novel approach to build a speech-to-speech conversion (STSC) system for individuals with speech impairment dysarthria is described. STSC system takes impaired speech having inherent disturbance as input and produces a synthesized output speech with good pronunciation and noise free utterance. The STSC system involves two stages, namely automatic speech recognition (ASR) and automatic speech synthesis. ASR transforms speech into text, while automatic speech synthesis (or text-to-speech [TTS]) performs the reverse task. At present, the recognition system is developed for a small vocabulary of 50 words and the accuracy of 94% is achieved for normal speakers and 88% for speakers with dysarthria. The output speech of TTS system has achieved a MOS value of 4.5 out of 5 as obtained by averaging the response of 20 listeners. This method of STSC would be an augmentative and alternative communication aid for speakers with dysarthria.

KEYWORDS

All India Institute of Speech and Hearing (AIISH), Artificial Neural Networks(ANN), Automatic Speech Recognition (ASR), Linear Predictive Coding (LPC), Speech-to-Speech Conversion (STSC), Text-to-Speech (TTS)

1. INTRODUCTION

Speech is the natural way of information flow in human beings. There are many barriers along the speech communication pathway. Among them, impairment in speech such as dysarthria is the major disorder affecting the normal life of a large number of individuals around the globe. This speech impairment occurs in 1.4% of the population. There are different clinical approaches in practice in treating speech disorders, but very less work is progressed using engineering technologies such as Artificial Intelligence. The present work focuses on individuals with speech impairment dysarthria, which is characterized by "irregularity of volume, pitch, resonance, which is ill-suited for the gender, age, or culture of the speaker". Common causes for dysarthria include throat muscle weakness, neurological disorders, brain injury, and brain tumors. Depending on the severity index, dysarthric speakers can be put in one of the three groups as mild, moderate, and severe. Speech utterances

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produced by dysarthria speakers can be very arduous for listeners unfamiliar with the speaker to perceive. Some recent research works have concentrated on the assessment of dysarthric speech and the use of automatic speech recognition (ASR) systems. Many researchers explained linear models to address the issues in assessing dysarthric speech such as, using normal speech as a control in evaluating dysarthric speech, classifying speech flaws in terms of its characteristic features, and finding the correlation between intelligibility rating and speech recognition performance (Dey, N., 2019). Also, few researchers examined computerized recognition of dysarthric speech using structured model approaches and concluded that the low overall recognition rate of dysarthric speakers remains a serious problem.

The main motivation of this work is - a very limited number of research works have focused on augmentative supports such as intelligible speech-generating devices for dysarthric speakers. It is observed that the recognition performance of the Automatic Speech Recognition (ASR) systems is considerably low. The proposed system describes an augmentative communication aid for speakers with dysarthria. The system presented in this paper aims to address the issue of low recognition performance observed with the previous ASR research works, develop a Text-to-Speech engine with intelligible speech and development of a speech-to-speech conversion device in the later stage. The significance of the proposed system lies in the fact that it would be an "augmentative *and alternative communication aid "for* dysarthric speakers.

The rest of the article is split into the following sections.

Section-2 deals with the literature review. Section-3 discusses the proposed system model and ASR. Section-4 provides the details of Kannada TTS engine. Section-5 deals with system evaluation, Section-6 presents results and discussion. Section-7 deals with the conclusion of the work.

2. BACKGROUND

A brief literature review is presented in this section. The criteria used to identify the articles to review are- existing ASR systems for recognizing disordered speech, ASR techniques (Nasereddin, H. O., Hebah & Omari, Ayoub, 2017) and development of Text-to-Speech conversion technique (Ravi, D.J, & Patilkulkarni, P., 2011).

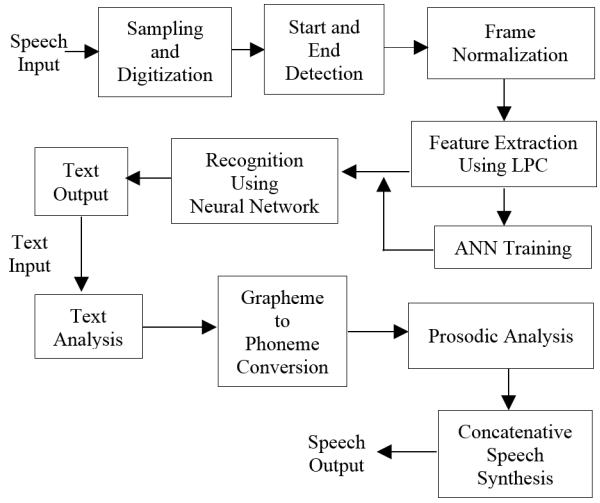
(Rosen K and Yampolsky, S., 2000) reviewed functioning of automatic speech recognition with dysarthric speech and found that the speaker-dependent speech recognition is suitable for severe dysarthria as it facilitates individuals to train the system with their own utterances instead of requiring speech that is near to 'normal'. "Recognition of command words from severely dysarthric speakers, who have poor control over their articulators" is proposed by (Green, P. *et al.*,2003). The described work utilizes a small wordbook (10 words) of isolated utterances for speaker-dependent application. For moderately dysarthric speech, the recognition performance of the speaker-dependent recognisers considering a small vocabulary is good (Polur, PD., Miller GE., 2005) but, as the vocabulary size increases performance for severely dysarthric speech is degraded (Hawley, MS et al., 2003). (Seyed Reza Shahamiri, Siti Salwah & Binti Salim,2014) have identified the best performing set of MFCC feature vectors, which can represent dysarthric acoustic feature parameters to be used in artificial neural network (ANN) based speech recognition. "The work studies the application of ANNs as a fixed-length isolated-word speaker-independent ASR for persons suffering from dysarthria". (Mendiratta, S., *et al*., 2015) proposed an ASR system employing Cuckoo Search Optimization (CSO) algorithm-based ANN Classifier. This method has discussed two kinds of feature extraction techniques MFCCs and Linear Predictive Coding Coefficients. The issue associated with the methods proposed was the low recognition rate. (Lee, T., *et al.*,2016) described the application of computerized speech recognition for objective assessment of voice and speech disorders. The study focused on Cantonese language and the issue with this approach was the high error rate in recognition. Dysphonia and Aphasia are the two disorders related to speech production were studied. (Gupta, S. *et al*.,2016) proposed a noise-robust ASR developed based on ANNs. The method used Mel frequency Cepstral Coefficients (MFCCs) as

feature vectors. Speech recognition accuracy obtained is impressive and it is observed that the ANN training algorithm developed has played a very important role in the system's performance. More recently, (Sen, S., et al., 2019) have discussed the overview of audio processing, it involves recent developments in the methodologies employed in audio processing and speech recognition. Also, authors have discussed the significance of audio indexing and presented two notable indexing methods, specifically Phonetic Search and Large Vocabulary Continuous Speech Recognition (LVCSR). The issues related to the human speech production system to produce artificial speech have been addressed. (Biadsy, et al., 2019) discussed a thoroughly trained speech-to-speech conversion system that maps an input spectrogram to a different spectrogram. The system described consisted of an encoder, spectrogram, and phoneme decoders, followed by a vocoder to generate speech waveform in time-domain. The system achieved success in impaired speech normalization. But the speaker's original voice identity has vanished in this process. (Dey, N., 2019) Investigated the use of speech signal data processing covering diverse systems and associated real-world activities and applications. The author has covered several areas such as biomedical speech recognition, convolutional neural networks, and speech disorder identification. (Yang, et al., 2019) proposed a conversion method which was based on Generative Adversarial Network (GAN) and was successful in preserving the speaker's voice in the foreign-accented speech domain. (Santosh, K. C., et al., 2019) compiled recent advancements in speech, audio and music signal processing. Various topics such as voice activity detection (VAD), speech coding methods, and wavelet filter banks for speech recognition applications were discussed. (Das, N., Chakraborty, S., Chaki, J. et al., 2020) presented a state-of-the-art overview of techniques for employing the extensively used machine learning and deep learning methods to identify the difficulties associated. Also, briefed about future research directions in improving the quality of speech.

(Ravi, D.J, and Patilkulkarni, P., 2011). The designed system has addressed issues such as linguistic analysis and customization of the synthesized speech. (Reddy, M.V. *et al*.,2014) proposed a work, which describes the basic details about the phonetics (classification of speech sounds) of the Kannada syllables and place of articulation; helpful in building the phonetic dictionary. (Chakraborty, K. *et al.,*2014) proposed algorithm for speech recognition using MFCCs as feature vectors. Euclidean distance is used to compare the template and real-time input. Results have shown good accuracy in voice recognition.

The literature review discussed above reveals that, though there is a significant improvement in ASR systems and TTS systems independently over the years, very little progress is observed in the development of alternative communication aid for speech disordered individuals. The proposed system would be an "augmentative and alternative communication aid such that the system would automatically recognize the individual's utterance and then convert the recognized speech to an acoustic signal". Thus, the system would address intelligibility issues due to Dysarthria (voice source degradation). The paper can be effectively considered in two phases namely, the first one is ASR and the second one is TTS. "ASR is the process in which the system perceives the meaning of the input acoustic signal. This work employs Linear Predictive Coding (LPC) for feature extraction of speech signals and ANN as a classifier. In Text-to-speech (TTS) process written text is converted into speech utterances. The process involves two phases: Text processing, and acoustic signal generation & processing. The text refining process converts the text into respective phonemic and syllable units. Speech generation involves the production of sounds matching to each phoneme and/or syllable. Irrespective of the method used, a substantial speech corpus is required for speech generation. Formant synthesis, articulatory synthesis, and concatenative synthesis are the computerized speech generation techniques used in the research practices. Articulatory speech synthesis requires modelling of the human speech production process. Source filter modelling is used in formant synthesis. Concatenative synthesis is used in this work, as the complexity involved in the development of synthesizer is less and can produce natural-sounding speech (Ravi, D.J, and Patilkulkarni, P., 2011 & Shreekanth, T. et al.,2014).

Figure 1. ANN-based isolated word ASR system integrated with Text-to-Speech system producing an STSC model



3. PROPOSED SYSTEM MODEL

The proposed STSC system model is illustrated in Figure 1. It can be noticed from the figure that the initial portions depict the ASR using ANN and later portions after the generation of recognized text, presents the TTS system.

3.1 Automatic Speech Recognition System

The isolated word recognition system presented has two phases namely 1) Feature extraction & Training and 2) Testing. The ASR system is implemented as shown in Figure 1.

3.1.1. Acquisition of Speech Signal

Recording of the speech is carried out using a standard noise cancellation type of microphone in a noise-free environment. Isolated word utterances are recorded and stored in .wav format. The noise present in the acquired signal is minimized through Blind Source Separation (Khosravy, M., et al., 2010 and Khosravy, M., et al., 2020). The Analog-to-Digital conversion is essential for digital processing of the signal & is normally carried out by the internal digital signal processor on a personal computer. The preferred sampling rate is 8000 Hz because the frequency range for human speech is approximated to be 300 to 3000 Hz. With 8000 Hz sampling frequency, The *Nyquist frequency* obtained is 4000 Hz, which is sufficient for a 3000 Hz speech signal. The encoding which can be used is 8 or 16 bits per sample, that the sound cards can achieve.

3.1.2. Pre-Emphasis Filtering

The purpose of pre-emphasis is to magnify the high-frequency components of the speech signal. Since the speech signal has a total spectral lean of 5 to 12 dB per octave, pre-emphasis filtering is used on the speech signal. As higher formants contain less energy when compared to lower formants, pre-emphasis filtering will compensate for the magnitude of high-frequency components.

3.1.3. Feature Extraction

Usually the features of speech signal for recognition systems are derived from Linear Predictive Coding (LPC) and MFCC's. This work has utilized the LPC method of feature extraction as the complexity involved in computation is less as compared to MFCCs computation. The LPC technique tries to

extract the coefficients of a filter that would generate the utterance which is being processed. Since, Linear predictive coding has the potential to derive and store time-varying formant information, which is useful in speech signal processing (Unnibhavi, A. H., & Jangamshetti, D. S.,2017). In LPC of speech, the nth sample of speech is calculated as a linear amalgamation of previous output speech samples. The LPC coefficients capture almost all formant frequencies and hence can effectively represent the voice samples through a lesser number of coefficients. The following section explains the calculation of LPC coefficients.

Considering $x(n)$ as the input signal and $\hat{x}(n)$ as the signal to be predicted, equation for Linear Predictive Coding is:

$$\hat{x}\left(n+1\right) = \sum_{l=0}^{P-1} w\left(l\right) x\left(n-l\right) \tag{1}$$

where, $\hat{x}\left(n+1\right)$ is the predicted value of next input sample $x\left(n+1\right)$ and $w\left(0\right)$ $w\left(1\right)\ldots w(p-1)$ are the weights multiplied for a 'p' length filter to get $\hat{x}\left(n+1\right)$.

Error in the prediction values is given by the equation (2):

$$E\left(n\right) = x\left(n+1\right) - \hat{x}\left(n+1\right) \tag{2}$$

The Mean Square Error is calculated by using equation (3):

$$\text{E}\left[\ \left|e\left(n\right)\right|^2\right] = E\left[e\left(n\right)e^*\left(n\right)\right] \tag{3}$$

where, $e^*\left(n\right)$ is the complex conjugate of the error.

To get more precise prediction, the mean square error should be reduced significantly. The minimised mean square error is given by the equation (4):

$$\text{Є} = \text{E}\left[\ e\left(n\right)\ \left\{\ x^*\left(n+1\right) - \sum_{l=0}^{p-1} w^*\left(l\right)\ x^*\left(n-l\right)\right\}\right] \tag{4}$$

As 'Є' is a real function of complex variables, it is differentiated with respect to $w^*\left(l\right)$ or $w\left(k\right)$ and equated to zero as:

$$\frac{\delta E}{\delta^* w^*\left(k\right)} = 0;\ \text{k=0, 1…p-1} \tag{5}$$

After differentiating equation (5):

$$\text{E}\left[\frac{\delta}{\delta w^*\left(k\right)}\left\{e\left(n\right)\left\{x^*\left(n+1\right)\right.\right.\right. - \frac{\delta}{\delta w^*\left(k\right)}\left\{e\left(n\right)\sum_{l=0}^{p-1} w^*\left(l\right)x^*\left(n-l\right)\right\}\Bigg] = 0 \tag{6}$$

Since none of the *w(k)* first term leads to zero, which results in:

$$E\left[-e\left(n\right)x^{*}\left(n-k\right)\right]=0$$

$$E\left[e\left(n\right)x^{*}\left(n-k\right)\right]=0;\text{ for k=0…p-1} \tag{7}$$

Substituting for $e\left(n\right)$ in equation (6):

$$\text{E }[\left\{x\left(n+1\right)-\sum_{l=0}^{p-1}w^{*}\left(l\right)\ x\left(n-l\right)\right\}\ x^{*}\left(n-k\right)\ =0$$

$$\text{E }\{x\left(n+1\right)x^{*}\left(n-k\right)\text{- }\sum_{l=0}^{p-1}w\left(l\right)\ \text{E}\left\{x\left(n-l\right)x^{*}\left(n-k\right)\right\}\ =0 \tag{8}$$

Assuming that the process to be wide sense stationary i.e.:

$$\text{E }\{\text{x }(k\ x^{*}\left(l\right)\}\ =\ r_{x}\,(k-l \tag{9}$$

Equation 8 reduces to:

$$r_{x}\ (k+1)\text{- }\sum_{l=0}^{p-1}w\left(l\right)\ r_{x}\ (k-l)=0;\text{ for k=0…p-1} \tag{10}$$

The auto-correlation matrix and hence the LPC coefficients generated is given by the equation (11):

$$\begin{pmatrix} r_{x}\left(1\right) \\ r_{x}\left(2\right) \\ \vdots \\ \vdots \\ r_{x}\left(p\right) \end{pmatrix}=\begin{pmatrix} r_{x}\left(0\right) & r_{x}^{*}\left(1\right)\ldots\ldots & r_{x}^{*}\left(p-1\right) \\ r_{x}\left(1\right) & r_{x}^{*}\left(0\right)\ldots\ldots & r_{x}^{*}\left(p-1\right) \\ & \vdots & \\ & \vdots & \\ & r_{x}\left(p-1\right)\ldots\ldots\ldots r_{x}\left(0\right) & \end{pmatrix}\begin{pmatrix} w\left(0\right) \\ w\left(1\right) \\ \vdots \\ \vdots \\ w\left(p-1\right) \end{pmatrix} \tag{11}$$

where, $w\left(0\right)\ \ w\left(1\right)\ldots w(p-1)$ are called as LPC coefficients. These LPC coefficients are used to train the neural network to classify the voice signals.

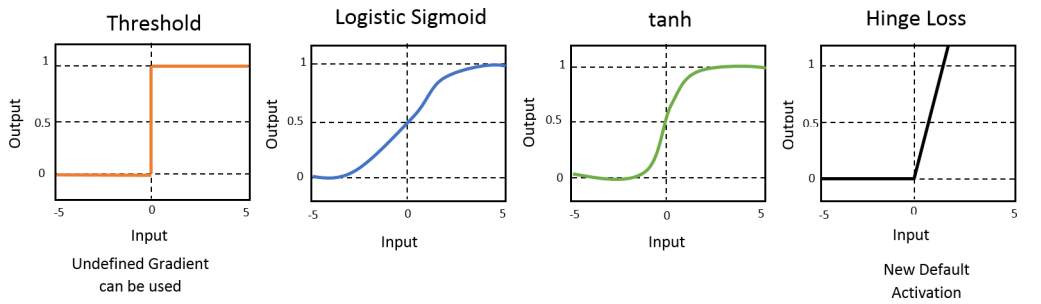
3.1.4. Recognition

The recognition component is fed with the LPC Coefficients extracted from the spoken utterances as input. This work has employed supervised ANNs as the recognizing unit. The recognizing component is explained in the following section.

3.2. Acoustic Model

A neural network system has the capability of recognizing the words for which it is trained. Word-based acoustic model is utilized for implementing a small vocabulary of 50 words. In the case of a large vocabulary implementation, phone-based models are employed.

Figure 2. Characteristic curves of the activation function



3.2.1. Activation Function

The neural network system is made up of a particular number of processing units that mimic neurons of the human brain. Each neuron that receives data, carry out a predefined function. The resultant value of this function is communicated with connected neurons. The sharing of resultant value is called the activation process and the value shared is activation value. The activation value determines whether the receiving neuron will be activated or to remain in the in-active state (James M. Keller et al.,2016). Different activation functions are used in practice for approximating the activation value to either 1 to 0 or 1 to -1. The most common activation functions available are represented in Figure 2 through their characteristic curves. The log-sigmoid activation function is been utilized for this work.

The mathematical form of log-sigmoid activation function is given by:

$$\sigma(x) = \frac{1}{1+e^{-x}} \tag{12}$$

3.2.2. Neural Network Structures

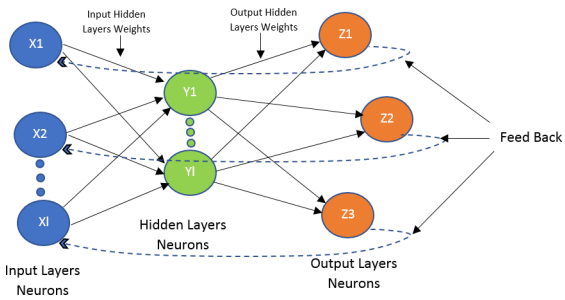
Artificial neural networks can be categorized in two significant groups, feedforward neural networks, and recurrent neural networks. *A* recurrent neural network is based on a *back-propagation algorithm* (James M. Keller et al.,2016 & Nassif, A. B., et al., 2019). Recurrent neural network is a directed and weighted graph that has at least one feedback loop as shown in Figure 3. This approach enables data to move backward across the neural network and adjust weights depending on updated outputs (Vieira, et al., 2017). With the output data value now being able to change weights in the network, they can exhibit a more effective temporal behavior.

3.3. Implementation

The ANN based ASR system is implemented on a general-purpose computer. The components used for implementing the ASR system are:

1. Standard noise cancellation Microphone.
2. Personal computer System.
3. The recognition component of the proposed STSC system is implemented using MATLAB tool (MATLAB.,2018). For the purpose of database creation, recording, and pre-processing of speech needs to be carried out. It is achieved using the NCH-Wave pad (NCH Wavepad., 2018) and PRAAT tools (Praat tutorial, 2018).

Figure 3. Structure of Recurrent Neural Network



3.4. Database Creation and Training of the Neural Network

A database is created by recording 50 isolated words in a noise-free room through different speakers, which consited of normal speakers and individuals with Dysarthria from different age groups and gender. The number of speakers chosen for training the neural network was 20 (10 male & 10 female) including 15 speech disordered individuals (Dysarthric speakers), aged between 15 to 65 years. Each word is recorded by every person 10 times (Gupta, S. *et al*.,2016). The LPC coefficients are obtained for each frame of the speech signal as explained in section-3.1.3.

3.4.1 Training

The LPC coefficients generated are used to train the neural network. Since the window length is chosen 30ms which accounts for a block length of 240 and the cross-correlation used at the later stage has +/- 12 lag, a neural network with 228 input layers (228 LPC coefficients for each word) is developed. 100 hidden layers i.e., less than half the size of input layers (Gnana Sheela, K., and Deepa, S. N., 2013), and 50 output layers are used in the neural network implemented. The structure of the same is depicted in Figure 5.

3.4.2 Testing

During the recognition phase, a spoken word is recorded with the same sampling rate of 8000 samples/sec. The signal is then processed to remove the silence by setting a threshold amplitude. Next, the filtering of the signal is carried out to minimize the noise, which is followed by the normalization of the signal. The LPC coefficients are calculated for this normalized signal and are input to the neural network for recognition.

To get accurate results, the number of training samples needs to be increased and the speech should be recorded in different acoustic conditions. Figure 4 shows the LPC trajectory of training speech signal for the word 'KARNATAKA', and Figure 6 shows the LPC trajectory of the test speech signal for the same word. The LPC coefficient's trajectories indicate that the linear prediction carried out produces a signal which is the closest approximation of the actual speech signal's envelope. The testing is carried out using MATLAB Graphical User Interface (GUI) developed (MATLAB.,2018), which allows users to click on a push button and speak the word to be recognized. The GUI also displays the recognized word in a text box in English text form. The GUI developed is as shown in Figure 7. The corresponding Kannada text of the recognized word is displayed in a separate plot as depicted in Figure 8.

4. TEXT-TO-SPEECH SYSTEM

In Section-3 speech recognition is described which is essentially speech to text conversion. Once the corresponding text of uttered word is generated, the next step is to produce the synthesized speech

Figure 4. LPC trajectory for training word 'KARNATAKA'

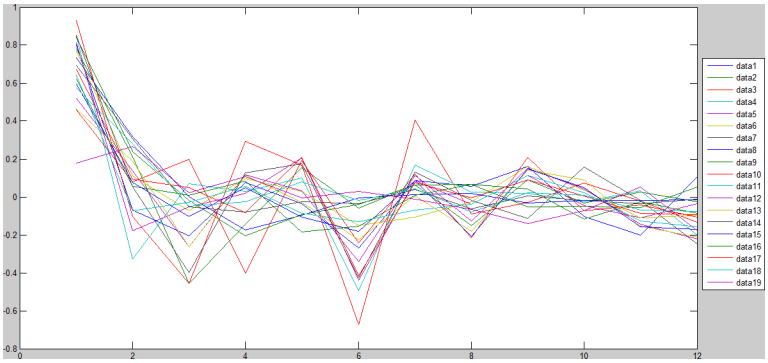


Figure 5. Structure of Neural Network

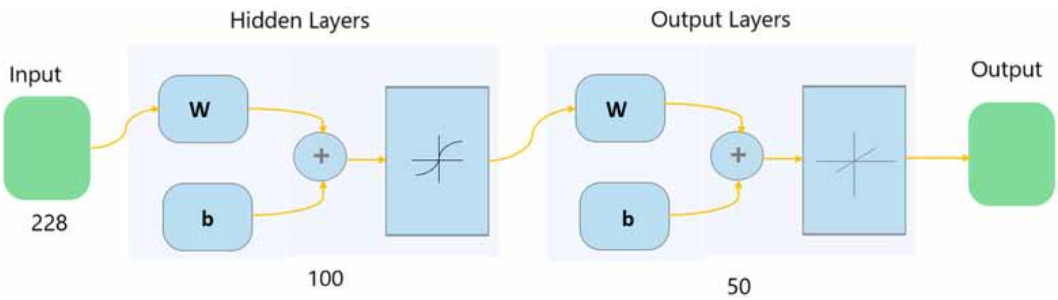
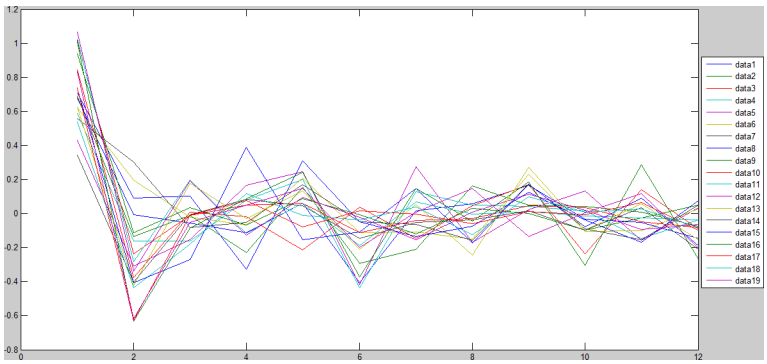


Figure 6. LPC trajectory of test signal 'KARNATAKA'



from the text. In TTS, the input text is split into individual letters, and respective phonic sounds are searched and concatenated to produce synthetic speech (Huang, X. et al.,2001 & Hlaing, C.S., 2017). The speech generated is played through the system's internal speaker. The following section discusses the procedure of TTS conversion for Kannada language.

4.1. Kannada Language

Kannada is the official language of Karnataka, a southern Indian state, and is spoken in the Indian subcontinent by over 60 million people. The Kannada character set is similar to that of other Indian

Figure 7. MATLAB GUI

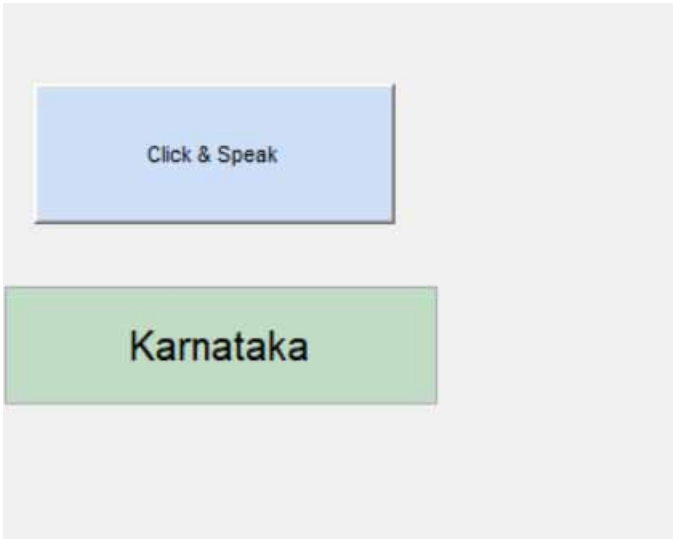


Figure 8. The recognized word 'KARNATAKA'

languages, as the script is phonetic in nature. "Each written symbol in Kannada script corresponds with one syllable, as compared to one phoneme in languages like English" (Kurian, A.S. et al., 2011). The language has 49 phonemic letters, classified into three groups: vowels (13 letters); consonants (34 letters); and neither vowel nor a consonant type of letters ('yogavaahakagalu': 'anusvara' ಂ and 'visarga' ಃ).

4.2. Unicode

Unicode is a universal character set. It provides a unique alpha-numeric number for each character in a language and supports all platforms and languages. These Unicode characters are essential in building the database and concatenation of phonemes (Srinath Sastry, C.V.).

4.3 Database Creation

During the speech synthesis operation, required phoneme units are searched and picked from the developed database, concatenated, and then processed suitably. Hence, creating a flawless database of phonemes is required. To meet these requirements, a utility tool called Praat is used. The phonemes were recorded with a sampling rate of 16000 samples/sec and the encoded by a scheme of 16 bits/sample. The sound and duration of the phonemes in any language varies slightly as a consequence of their occurrence in the speech.

The different positions at which phonemes can occur are:

1. **Initial:** At the beginning of a word.
2. **Middle:** In between two phonemes.
3. **Final:** At the end of the word.

To get naturally sounding speech at the output of TTS, each phoneme unit is fetched from a word/phrase involving the phoneme. Three words/phrases were picked such that, they have essential phonemes at the initial, middle and final positions. Then, the phonemes were extracted from the specific position and saved in directories named initial, middle, and final with a unique numerical name corresponding to their Unicode (Ravi, D.J, and Patilkulkarni, P., 2011).

Database building steps:

1. Open the Praat utility tool. Choose record 'mono' sound option from 'new' icon in the menu bar.
2. A sampling frequency of 16000 Hz is chosen and press 'record' icon to start recording.
3. Speak out the word/phrase having needed phoneme.
4. After recording the word/phrase stop recording and save it to a list.
5. Use the 'Edit' option to display the waveform, search carefully for the required phoneme in the waveform and extract that portion and save with unique numerical name in .wav format.
6. Let the phoneme be ಕ. Three Kannada words ಕಮಲ, ಮಕರ, and ತುಲಕ are recorded. From each of the recorded word ಕ is extracted and stored as a different '.wav' files in different directories. The phonemes are extracted as:
   ಕಮಲ – ಕ in the initial position
   ಮಕರ – ಕ in the middle position
   ತುಲಕ – ಕ in the final position

The phoneme is saved in '.wav' format with the unique numeric name, in this case, the name of the file is 322100.wav. The file name of the above example is generated as per Table 2 and Table 4. The rich database developed for the Kannada language consists of total 1605 phonemes. Each position has 535 phonemes. The recorded phonemes comprise consonants, vowels, semivowels, stops, fricatives, nasals, etc.

4.4 Pre-Processing Stage

Prior to synthesizing, preprocessing of the text is required. The primary intention of the pre-processing stage is to fix the ambiguities existing in the text. To resolve ambiguities, the Kannada alphabets and symbols are grouped into different classes as shown below in Table 1 to Table 4. The Decimal equivalent Unicodes of the character set is also shown in the corresponding table. It can be observed that Table 4 indicate digits to be padded to the actual Unicodes of consonants to generate Unicode of compound characters. "The pre-processing algorithm scans the input text, reads it character by character (Unicode) and produces a reformed Unicode file". The reformed Unicode file is then passed to an algorithm developed using MATLAB tool for synthesizing the speech.

The parser reads the modified Unicode file, number by number and picks the respective phonemes from the database developed. The phoneme directory (i.e. Initial, middle or final needs to be selected while fetching phonemes is determined by examining the presence of spaces in the Modified Unicode file. After fetching the phonemes, the following steps are followed to obtain the synthesized speech:

1. The fetched phonemes are concatenated.
2. Moving average hamming window is used to combine the phonemes at their boundaries.
3. To smoothen the synthesized artificial speech, a speech enhancement technique is used. In this technique, the high-frequency component of synthesized speech is minimized and the low pass

component is added to spectrum and processed speech is fed to the loudspeaker. Thus, the speech to speech conversion process is completed.

Similarly, the other consonants are considered for the groups 03,04,05,06 and 07 (Refer to the APPENDIX).

5. SYSTEM EVALUATION

Evaluation of the STSC system involves ASR system testing followed by Validation of the TTS system through Mean Opinion Scores (MOS) and Dynamic Rhyme Tests (DRT).

Table 1. Group-01 Independent vowels

Kannada Phonemes	Unicode
ಅ	3205
ಆ	3206
ಇ	3207
ಈ	3208
ಉ	3209
ಊ	3210
ಋ	3211
ಎ	3214
ಏ	3215
ಐ	3216
ಒ	3218
ಓ	3219
ಔ	3220
ಂ	3202
ಃ	3203

Table 2. Group-02 Consonants

Kannada Syllables	Unicode
ಕ	3221
ಖ	3222
ಗ	3223
ಘ	3224
ಙ	3225

Table 3. Group-08 Dependent vowel signs

Dependent Vowel Signs	Unicode
ั	3277
า	3262
ิ	3263
ี	3264
ึ	3265
ื	3266
ุ	3267
ู	3270
ฺ	3271
เ	3272
แ	3274
โ	3275
ใ	3276

Table 4. Padding table

Dependent Vowel Signs	Digit to be Padded
ั	00
า	01
ิ	02
ี	03
ึ	04
ื	05
ุ	06
ู	07
ฺ	08
เ	09
แ	10
โ	11
ใ	12
๐	13
ๆ	14

5.1 ASR System Testing

In the first phase, the ASR system was tested for recognizing 50 different words with normal speakers aged between 15 - 65 years (5 speakers including 2 female) with good pronunciation ability. In the second phase, testing is done for the same 50 words uttered by 5 persons (including 2 female speakers)

having speech disorder- Dysarthria. The results of the first and second phases of testing are presented in Table 5. The accuracy mentioned in the tables is calculated using the *'Confusion Matrix'* (Confusion matrix, 2019). The confusion matrix is basically a 2*2 table that deals with four outcomes produced by a binary classifier. The four outcomes of the confusion matrix are:

1. True-positive (TP): correct positive recognition
2. False-positive (FP): incorrect positive recognition
3. True-negative (TN): correct negative recognition
4. False-negative (FN): incorrect negative recognition

In the context of this work, if the input word is recognized correctly then the result of the test is true positive. If the speech input is correct and recognized word is incorrect then, the result of the test is true negative. Using confusion matrix, Accuracy (ACC) is calculated as the ratio of the sum of all correct recognitions to the total dataset available. It is given by the equation:

$$\text{ACC} = \frac{TP + TN}{P + N} \tag{13}$$

where $P + N$ indicates the total number of data set.

For accuracy calculation, true positive (TP) & true negative (TN) outcomes were used. It can be observed from Table 5, that the accuracy is least when hidden layers are less and it becomes constant for a larger number of hidden layers. However, as the number of hidden layers utilized increased beyond 130, the accuracy starts decreasing. This drop-in accuracy is due to the fact that, as number of hidden layers increases, the neural network starts creating un-necessary relations among the data and thus recognizing the wrong word.

5.2. Validation of the STSC System

For the purpose of validation of STSC, the isolated words were spoken and corresponding synthetic speech is generated. The artificial speech generated is played in front of an audience using a standard multimedia PC with internal speakers. The perceiver's opinion is recorded for evaluation of the output and the Mean Opinion Score (MOS) for every word is calculated. "MOS is expressed as a single number in the range 1 to 5. Where 5 refers to the highest perceived quality and 1 refers the lowest perceived quality. For acoustic signals, MOS tests are defined by ITU-T" (Waidyanath, N., 2012).

In the context of this work, 50 words listed in Table 6 are used for training the ASR system's neural network. Testing the intelligibility and naturalness is done for the synthesized speech. The average MOS rating is calculated for each word based on MOS ranges assigned by the listeners as tabulated in Table 6. The average MOS score achieved is 4.5 out of 5 indicating that the synthesized speech is natural sounding and easily perceivable.

"Diagnostic Rhyme Test (DRT) is carried out to measure the intelligibility of the STSC system". The DRT has provided promising results, as negligible mismatches were observed between recognized speech and synthesized speech output. Figure 9 shows the waveform of spoken word and Figure 10 shows the synthesized sound waveform with the recognized text for the word KARNATAKA (ಕರ್ನಾಟಕ).

It can be observed from the figures (Figure 9 & Figure 10) that the waveform of synthesized speech is the close approximation of input speech signal with respect to magnitude, but the duration of computerized speech is more than that of the input-indicating the time gap between successive concatenation of phonemes. Precise phoneme database creation and concatenation will minimize this limitation.

Table 5. Recognition accuracy of 50 different words with the input feature vector of length 228 for normal speakers with good pronunciation ability and Individuals suffering from Dysarthria

No. of Neurons in Hidden Layer	Accuracy of Normal Speakers	Accuracy of Disordered Speakers
10	72%	52%
20	80%	70%
30	82%	74%
40	86%	76%
50	90%	80%
60	92%	82%
70	92%	84%
80	94%	86%
90	94%	86%
100	94%	88%
110	94%	88%
120	94%	88%
130	94%	88%
140	92%	86%
150	90%	82%

6. RESULTS AND DISCUSSION

The training and testing of the system are carried out in a noise-free environment and MATLAB tool is employed for simulating the stated process. The highest recognition rate that is achieved with the ASR system (first stage of STSC) is 94% for normal speakers and 88% for speakers with Dysarthria for 50 isolated words of Kannada language. This result is achieved employing ANN with a learning rate of the training process as 0.7. ANN is trained using 228 LPC coefficients per word. The minimum number of hidden layers required for maximum accuracy (94%) are 80 for normal speakers and 100 (for 88% accuracy) for speakers with Dysarthria. Hidden layers are considered for the discussion, as the number of hidden layers is directly proportional to the computational delay. The result of the TTS system is evaluated to check for the naturalness and intelligibility of the synthesized speech on the developed database. Naturalness corresponds to the similarity to the human voice and the intelligibility corresponds to the ability to understand the synthesized speech. The quality of speech produced by TTS is excellent as validated by MOS (4.5 out of 5).

The proposed STSC system can be compared to Hidden Markov Models (HMMs) based Voice-Input Voice-Output Communication Aid (VIVOCA) developed by (Hawley et al., 2006) The device developed had issues in real time usage and has shown a mean recognition accuracy of 67%. While the mean accuracy of the proposed STSC system is very promising (88%), it would be an augmentative and alternative communication aid for Dysarthric speakers. Since the system has inbuilt TTS, which can be easily modified to use as a text-to-speech converter for visually challenged people & aged people suffering from diminished sight to read the Kannada text. The shortcomings observed in the system include-the system is developed for a vocabulary size of 50 isolated words which would not be sufficient to be a viable and useful aid for the communication. To overcome these limitations is the priority of our succeeding research.

Table 6. Test words and MOS rating

Sr. No	Kannada text	Pronunciation	MOS	Sr. No	Kannada text	Pronunciation	MOS
1	ಡಾ॥	Doctor	4.75	26	ನಿನ್ನ	Ninna	4.40
2	ಶ್ರೀಕಾಂತ್	Shreekanth	4.35	27	ಹೆಸರು	Hesaru	4.50
3	ಸುದರ್ಶನ್	Sudarshan	4.50	28	ಏನು	Yenu	4.55
4	ಸಿದ್ಧಣ್ಣ	Siddhanna	4.45	29	ನಿಮ್ಮ	Nimma	4.30
5	ಚಂದನ್	Chandan	4.60	30	ಊರು	Ooru	4.70
6	ಆದಿತ್ಯ	Adithya	4.75	31	ಧಾರವಾಡ	Dharavaada	4.25
7	ಅಭಿಷೇಕ್	Abhishek	4.25	32	ಜಾಸ್ತಿ	Jaasti	4.45
8	ಶ್ರೀ	Shree	4.35	33	ಕಮ್ಮಿ	Kami	4.45
9	ಜಯಚಾಮ	Jayachama	4.75	34	ಎಷ್ಟು	Yesthu	4.50
10	ರಾಜೇಂದ್ರ	Rajendra	4.25	35	ಘಂಟೆಗೆ	Ganthege	4.50
11	ತಾಂತ್ರಿಕ	Tantrika	4.25	36	ಕರ್ನಾಟಕ	Karnataka	4.60
12	ಮಹಾವಿದ್ಯಾಲಯ	Mahavidyalaya	4.45	37	ಇದೆ	Ide	4.75
13	ಮೈಸೂರು	Mysuru	4.35	38	ಇಲ್ಲ	Illa	4.75
14	ಗಂಗೋತ್ರಿ	Gangothri	4.40	39	ತಿಂಡಿ	Tindi	4.60
15	ಬಾಗಿಲು	Baagilu	4.75	40	ಇನ್ನು	Innu	4.55
16	ಕಿಟಕಿ	Kitaki	4.50	41	ಮಹೇಂದ್ರ	Mahendra	4.35
17	ಕನ್ನಡಿ	Kannadi	4.35	42	ವಿಶೇಷ	Vishesha	4.70
18	ಪುಸ್ತಕ	Pustaka	4.25	43	ಸಮಾಚಾರ	Samachara	4.70
19	ಲೇಖನಿ	Lekhani	4.60	44	ಗೊತ್ತಿಲ್ಲ	Gotthilla	4.45
20	ಪೆನ್ನು	Pennu	4.50	45	ಕಿವಿ	Kivi	4.75
21	ಶುಭೋದಯ	Shubodaya	4.30	46	ಗಣಪತಿ	Ganapathi	4.70
22	ಬೆಳ್ಗೆಗೆ	Beligge	4.40	47	ಆರಮನೆ	Aramane	4.70
23	ಮಧ್ಯಾಹ್ನ	Madyana	4.40	48	ಹೊಗೇಣಿ	Hogona	4.75
24	ಸಂಜೆ	Sanje	4.45	49	ಬರ್ತೀಯ	Barthiya	4.35
25	ವಿಜಯನಗರ	Vijayanagara	4.35	50	ಕನಸು	Kanasu	4.65

7. CONCLUSION

This paper has described the development of the STSC system. The system addresses the intelligibility issues of Dysarthric speech. The Evaluation of the system showed that the accuracy of the ASR component (88%) for disordered speech with a vocabulary size of 50 words is better when compared with the previous works (Cohen, A., Polur PD., Hawley & Lee, T.) However, the evaluation has also highlighted that the present work is limited in its capability, as the system can recognize 50 isolated words only. Further, the present system with a personal computer and head mount microphone makes the system not ideal to use. Feedback from the participants has confirmed that the system would be ideal if it can recognize continuous speech (at-least short messages) with improved accuracy. Also, it is noted that further research is required to transform the developed STSC system to a portable

Figure 9. Waveform of Input word 'KARNATAKA'

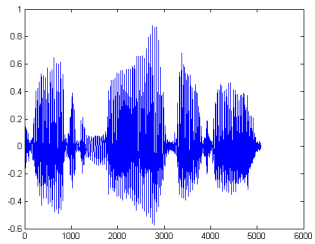
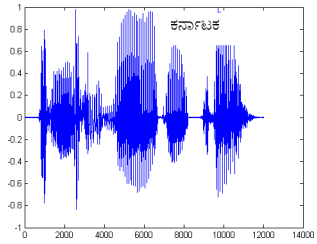


Figure 10. The synthesized word 'KARNATAKA' with the recognized text



device such that it can be affordable and acceptable alternative communication tool for Dysarthric speakers. The authors' future work is focused on overcoming these issues.

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APPENDIX

Kannada Consonant Groups and Corresponding Unicodes

Table 7. Group-03 Consonants

Kannada Syllables	Unicodes
ಚ	3226
ಛ	3227
ಜ	3228
ಝ	3229
ಞ	3230

Table 8. Group-04 Consonants

Kannada Syllables	Unicodes
ಟ	3231
ಠ	3232
ಡ	3233
ಢ	3234
ಣ	3235

Table 9. Group-05 Consonants

Kannada Syllables	Unicodes
ತ	3236
ಥ	3237
ದ	3238
ಧ	3239
ನ	3240

Table 10. Group-06 Consonants

Kannada Syllables	Unicodes
ಪ	3242
ಫ	3243
ಬ	3244
ಭ	3245
ಮ	3246

Table 11. Group-07 Consonants

Kannada Syllables	Unicodes
ಯ	3247
ರ	3248
ಱ	3249
ಲ	3250
ವ	3253
ಶ	3254
ಷ	3255
ಸ	3256
ಹ	3257
ಳ	3251

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